University of Asia Pacific

Department of Computer Science and Engineering

Course Code : CSE 450 (VLSI Lab)

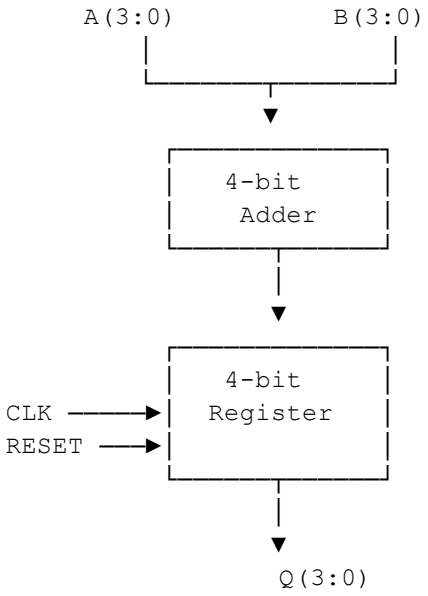
Mid-Term Lab Examination | Group A | Spring 2026

Duration	1 hr	**Total Marks**	20
Student Name	_____	**ID**	_____

Mid-Term Task: 4-bit Accumulator

Objective. Design and implement a 4-bit accumulator in VHDL by combining the 4-bit adder and register concepts developed in the laboratory. The system must perform addition synchronously with the clock and store the result in a 4-bit register.

System Architecture



Functional requirement. At every rising edge of CLK, the register stores the sum of the two 4-bit inputs. RESET has priority over the addition operation.

Required Operation

RESET	CLK	A	B	Operation / Q
1	↑	XXXX	XXXX	Q ← 0000
0	↑	A	B	Q ← A + B
0	No rising edge	XXXX	XXXX	Q remains unchanged

Required Test Sequence

Your testbench must produce the following sequence. The expected Q value should be observed after the corresponding rising clock edge.

Step	RESET	A	B	Operation	Expected Q	Explanation
1	1	0000	0000	Reset	0000	Register is cleared
2	0	0011	0101	3 + 5	1000	Load sum
3	0	0010	0001	2 + 1	0011	Load new sum
4	0	1111	0001	15 + 1	0000*	4-bit overflow
5	0	1010	0101	10 + 5	1111	Load new sum

*** Note:** For Step 4, the mathematical result is 16, but a 4-bit output can represent only 0–15. Therefore, Q = 0000 and the carry-out is discarded unless your implementation explicitly exposes it.

Implementation Requirements

- Use VHDL and IEEE.STD_LOGIC_1164.ALL.
- The register operation must be synchronized with the rising edge of CLK.
- RESET must be synchronous and active-high.
- Do not use a behavioral integer-based accumulator as a replacement for the required adder/register structure.
- The design should be modular and clearly identify the adder and register portions.
- The submitted testbench must generate a clock and apply the required input sequence.

Submission

While submitting to the classroom prepare the report from the third page of this template including the current template in the first and second page. From the third page write your report including but not limited to the following points.

- Attach your entity code and testbench code
- Attach your output visuals
- Challenges faced during the experiment
- Conclusion